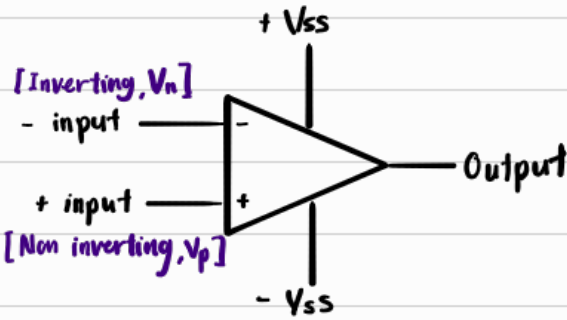


Operational Amplifiers (Op-Amp)



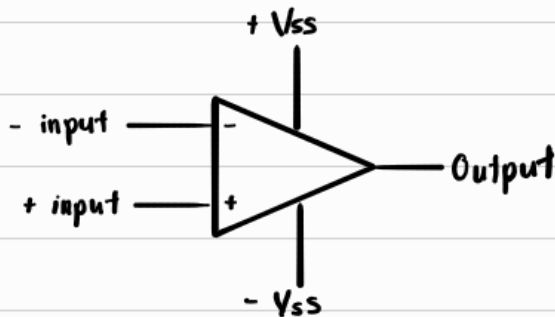
For ideal Op-Amp

- Infinite input impedance [$i_{input} = 0$]
- Zero output impedance [No "internal impedance"]
- Infinite gain, $a = \infty$
- Infinite bandwidth

An Op-Amp drives power directly from its supply pins to the output, scaled according to the voltage difference btwn its input pins.

$\therefore$ Output is limited by the supply $\Rightarrow V_{out} \leq V_{supplied}$.

Open Loop



$$\text{Open Loop Voltage Gain, } G_o = \frac{V_{out}}{(V_+ - V_-)}$$

This typically = 10^5

Negative Feedback

[Closed Loop]

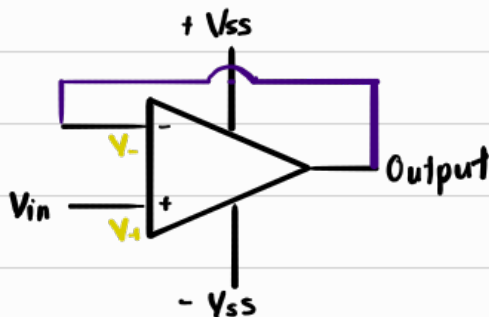
[Fllw Input]

[Voltage Fllwer / Unity - Gain]

V_{in}

In neg. feedback config. $\Rightarrow V_+ = V_- = V_{out}$

This happens cuz $V_{out} = G_o (V_+ - V_-)$

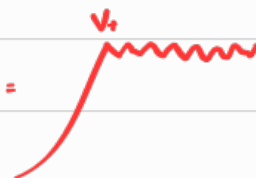


Loop

- Since G_o is huge [typically 10^5],
E.g. 0.001 when $V_{in} \approx 0$, V_{out} will still be huge.
- This pulls up V_{out}
- When $V_{out} = V_+ \Rightarrow V_- = V_+ \Rightarrow V_{in} = 0$
 $\hookrightarrow V_{out} = 0$
- This pulls down V_{out}

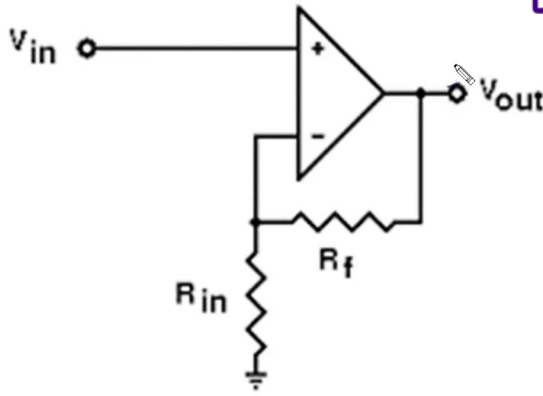
$$\text{Closed Loop Gain, } A = \frac{V_{output}}{V_{in}}$$

$\therefore V_{out} =$



Non-inverting amplifier

[Input at positive pin, V_+]



$$I_f = \frac{V_{in} - 0}{R_{in}}$$

$$I_f = \frac{V_{out} - V_{in}}{R_f}$$

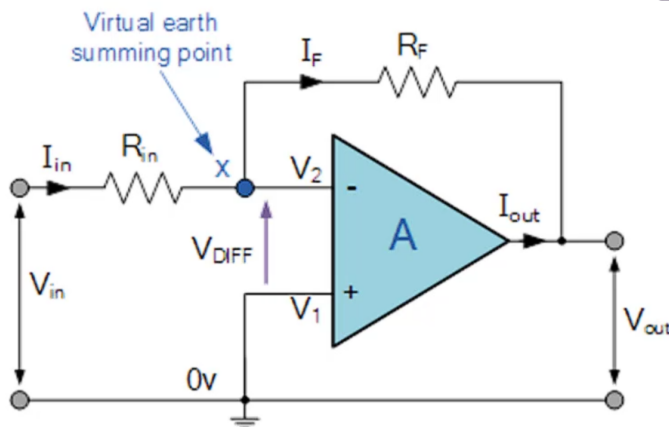
$$\therefore \frac{V_{in}}{R_{in}} = \frac{V_{out} - V_{in}}{R_f}$$

$$\frac{R_f}{R_{in}} = \frac{V_{out} - V_{in}}{V_{in}} = \frac{V_{out}}{V_{in}} - \frac{V_{in}}{V_{in}} = \frac{V_{out}}{V_{in}} - 1$$

$$A_v = \frac{V_{out}}{V_{in}} = 1 + \frac{R_f}{R_{in}}$$

Inverting amplifier

[Input at negative pin, V_-]



KCL at X: $I_{in} = i_- + I_f = 0 + I_f$

$$\frac{V_{in} - 0}{R_{in}} = \frac{0 - V_{out}}{R_f}$$

$$\frac{R_f}{R_{in}} (V_{in}) = (-V_{out})$$

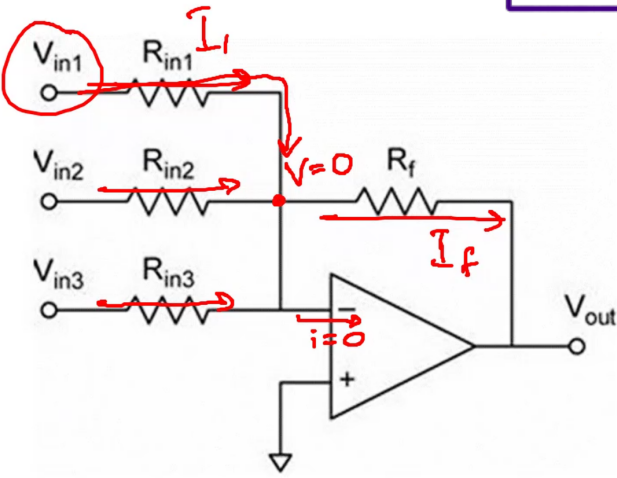
$$\frac{R_f}{R_{in}} = \frac{-V_{out}}{V_{in}}$$

$$\therefore \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_{in}}$$

$$\therefore A_v = \frac{V_{out}}{V_{in}}$$

$$A_v = \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_{in}}$$

Summing inverting amplifier



$$I_1 + I_2 + I_3 = I_f$$

$$\frac{V_{in1}}{R_{in1}} + \frac{V_{in2}}{R_{in2}} + \frac{V_{in3}}{R_{in3}} = \frac{0 - V_{out}}{R_f}$$

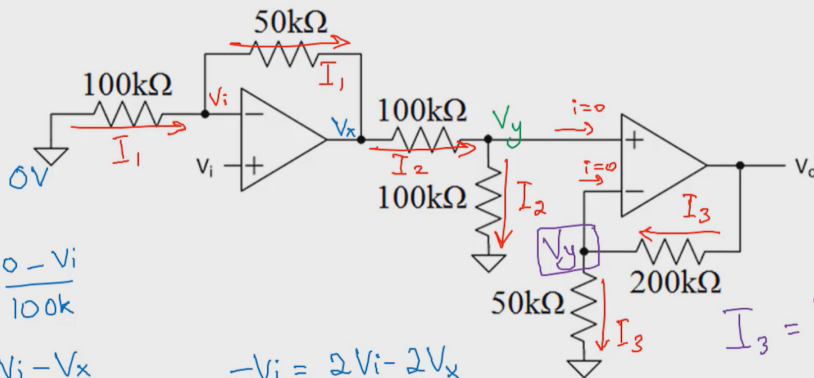
$$-\left[\frac{R_f}{R_{in1}} V_{in1} + \frac{R_f}{R_{in2}} V_{in2} + \frac{R_f}{R_{in3}} V_{in3} \right] = V_{out}$$

$$V_{out} = -\left[\frac{R_f}{R_{in1}} V_{in1} + \frac{R_f}{R_{in2}} V_{in2} + \frac{R_f}{R_{in3}} V_{in3} \right]$$

Example 8.2

$$A_v = \frac{V_o}{V_i}$$

Find the closed-loop gain of the following circuit:



$$I_1 = \frac{0 - V_i}{100k}$$

$$I_1 = \frac{V_i - V_x}{50k}$$

$$\frac{-V_i}{100k} = \frac{V_i - V_x}{50k}$$

$$-V_i = 2V_i - 2V_x$$

$$-3V_i = -2V_x$$

$$\therefore V_x = \frac{3}{2} V_i$$

$$I_2 = \frac{V_x - V_y}{100k} = \frac{V_y - 0}{100k}$$

$$V_x - V_y = V_y$$

$$V_x = 2V_y$$

$$V_y = \frac{1}{2} V_x$$

$$= \frac{1}{2} \left(\frac{3}{2} \right) V_i$$

$$= \frac{3}{4} V_i$$

$$I_3 = \frac{V_o - V_y}{200k} = \frac{V_y - 0}{50k}$$

$$V_o - V_y = 4V_y$$

$$V_o = 5V_y = 5 \left(\frac{3}{4} V_i \right) = \frac{15}{4} V_i$$

$$A_v = \frac{V_o}{V_i} = \frac{15}{4}$$

Differential amplifier

$$I_1 = \frac{V_1 - V_x}{R_1} = \frac{V_x - V_{out}}{R_f}$$

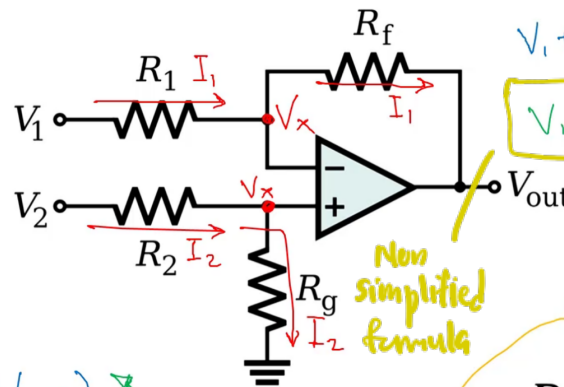
$$I_2 = \frac{V_2 - V_x}{R_2} = \frac{V_x - 0}{R_g}$$

$$V_1 - V_x = \frac{R_1}{R_f} (V_x - V_{out})$$

$$V_2 - V_x = \frac{R_2}{R_g} (V_x)$$

$$V_2 = \left(\frac{R_2}{R_g} + 1 \right) V_x = \frac{R_2 + R_g}{R_g} V_x$$

$$V_x = \frac{R_g}{R_g + R_2} V_2$$



$$V_1 + \frac{R_1}{R_f} V_{out} = \left(\frac{R_1}{R_f} + 1 \right) V_x$$

$$V_1 + \frac{R_1}{R_f} V_{out} = \left(\frac{R_1}{R_f} + 1 \right) \left(\frac{R_g}{R_g + R_2} \right) V_2$$

$$V_1 + \frac{R_1}{R_f} V_{out} = \left(\frac{R_1 + R_f}{R_f} \right) \left(\frac{R_f}{R_f + R_1} \right) V_2$$

$$V_1 + \frac{R_1}{R_f} V_{out} = V_2$$

$$\left. \begin{array}{l} R_g = R_f \\ R_2 = R_1 \end{array} \right\} \text{simplified formula}$$

$$V_{out} = \frac{R_f}{R_1} (V_2 - V_1)$$

Example 8.4

In the Differential amplifier, $V_1 = 2V$ and $V_2 = 3V$, and resistors $R_1 = 2k\Omega$, $R_2 = 6k\Omega$, $R_f = 6k\Omega$, and $R_g = 9k\Omega$.

a) What is the voltage measured at V_{out} ?

b) What should be the values of R_2 and R_g so that $V_{out} = \frac{R_f}{R_1} (V_2 - V_1)$?

$$V_1 + \frac{R_f}{R_1} V_{out} = \left(\frac{R_1}{R_f} + 1 \right) \left(\frac{R_g}{R_g + R_2} \right) V_2$$

$$2 + \frac{6k}{2k} V_{out} = \left(\frac{2k}{6k} + 1 \right) \left(\frac{9k}{9k + 6k} \right) (3)$$

$$2 + 3V_{out} = \left(\frac{8}{6} \right) \left(\frac{9}{15} \right) (3) = \frac{72}{30} = \frac{12}{5}$$

$$3V_{out} = \frac{12}{5} - 2 = \frac{12}{5} - \frac{10}{5} = \frac{2}{5}$$

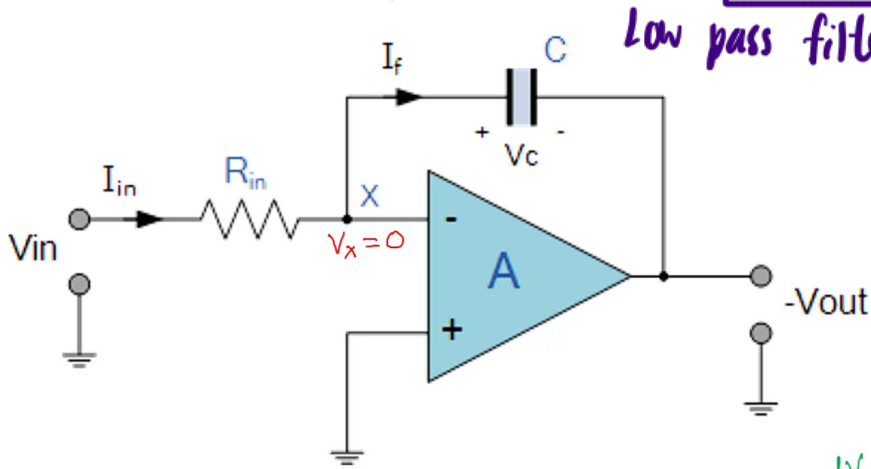
$$\therefore V_{out} = \frac{2}{15} V \neq$$

$$R_2 = R_1 = 2k\Omega \quad \#$$

$$R_g = R_f = 6k\Omega$$

Op-Amp Integrator

Low pass filter



$$A_v = - \left(\frac{1}{j\omega C} \right) \frac{1}{R_{in}} = \frac{-j}{2\pi f RC}$$

$$A_v \propto \frac{1}{f}$$

$f \uparrow$	$A_v \downarrow$
$f \downarrow$	$A_v \uparrow$

$$I_f = I_{in}$$

$$\frac{V_x - V_{out}}{Z_c} = \frac{V_{in} - V_x}{R_{in}}$$

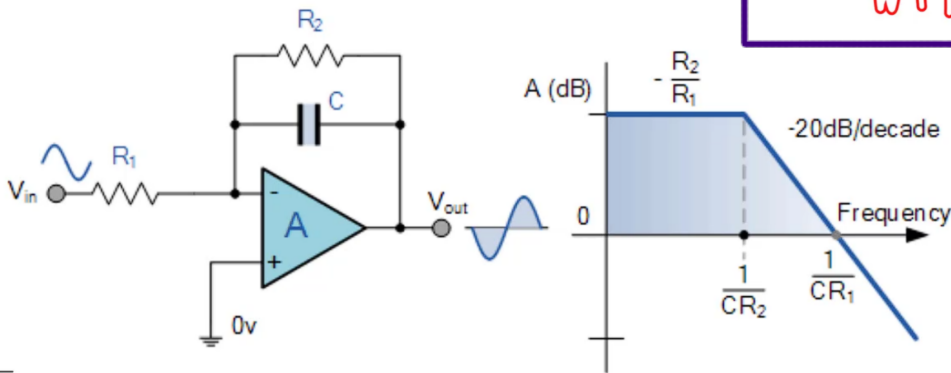
$$A_v = \frac{V_{out}}{V_{in}} = - \frac{Z_c}{R_{in}}$$

$$I_{in} = I_c = I_f = C \frac{dV_c}{dt} = C \frac{d(-V_{out})}{dt}$$

$$\frac{V_{in}}{R_{in}} = -C \frac{dV_{out}}{dt}$$

$$V_{out} = -\frac{1}{RC} \int V_{in} dt + V_{in}(0)$$

Op-Amp Integrator with DC gain



$$A_v = \frac{V_{out}}{V_s} = - \frac{Z_{R_2} \parallel Z_C}{Z_{R_1}}$$

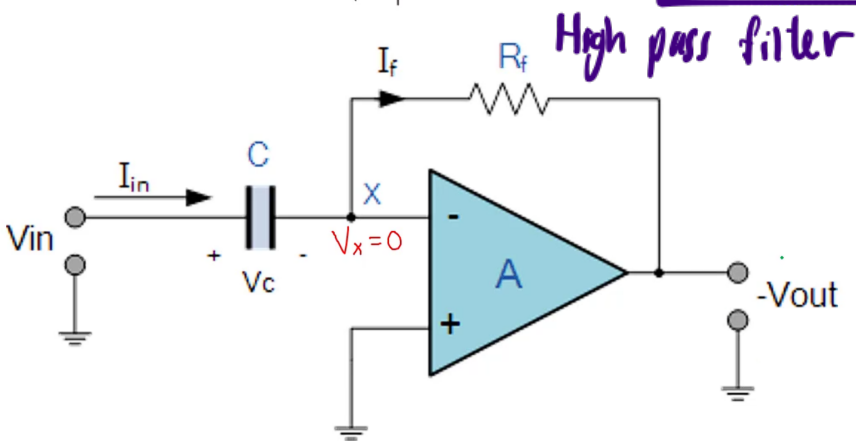
$$A_v = - \frac{R_2}{R_1} \frac{1}{1 + j\omega CR_2}$$

$$\angle A_v = 0 - \tan^{-1}(\omega CR_2 / 1)$$

$$\angle A_v = -\tan^{-1}(\omega CR_2)$$

$$|A_v| = \frac{R_2}{R_1} \sqrt{\frac{1}{1 + (\omega CR_2)^2}}$$

Op-Amp Differentiator



$$A_v = -\frac{R_f}{\left(\frac{1}{j\omega C}\right)} = -j\omega CR_f$$

$$A_v \propto f$$

$$\begin{matrix} f \uparrow & A_v \uparrow \\ f \downarrow & A_v \downarrow \end{matrix}$$

$$I_c = I_f = C \frac{dV_c}{dt} = C \frac{dV_{in}}{dt}$$

$$\frac{-V_{out}}{R_f} = C \frac{dV_{in}}{dt}$$

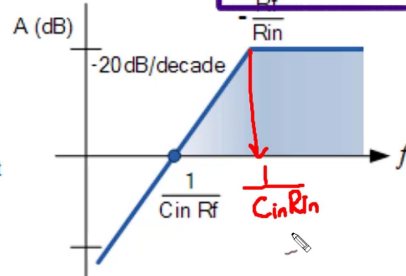
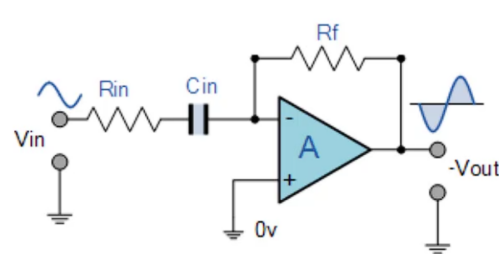
$$V_{out} = -RC \frac{dV_{in}}{dt} *$$

$$I_{in} = I_f$$

$$\frac{V_{in} - 0}{Z_c} = \frac{0 - V_{out}}{R_f}$$

$$A_v = \frac{V_{out}}{V_{in}} = -\frac{R_f}{Z_c}$$

Op-Amp Differentiator with DC gain



$$A_v = \frac{V_{out}}{V_s} = -\frac{R_f}{R_{in} + Z_c} = -\frac{R_f}{R_{in} + \frac{1}{j\omega C}}$$

$$A_v = -\frac{j\omega CR_f}{j\omega CR_{in} + 1}$$

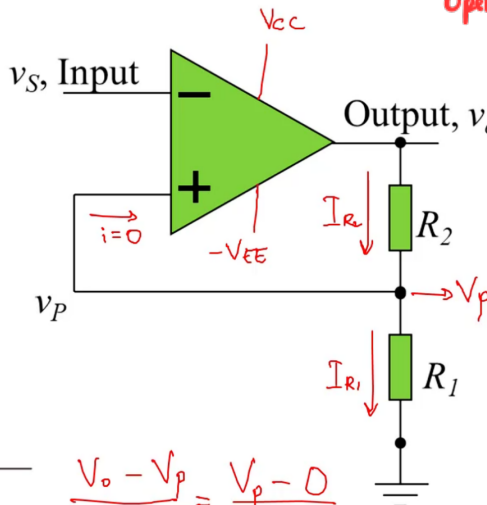
$$\angle A_v = \frac{\tan^{-1}(-\omega CR_f/0)}{\tan^{-1}(-\omega CR_{in}/-1)}$$

$$\angle A_v = -90^\circ - \tan^{-1}(\omega CR_{in})$$

$$|A_v| = \frac{\sqrt{(\omega CR_f)^2}}{\sqrt{(\omega CR_{in})^2 + 1}} = \frac{\omega CR_f}{\sqrt{(\omega CR_{in})^2 + 1}}$$

Schmidt Trigger

Open loop gain, $G_o = \frac{V_{out}}{(V_+ - V_-)} \Rightarrow V_{out} = G_o (V_+ - V_-)$



$$\frac{V_o - V_P}{R_2} = \frac{V_P - 0}{R_1}$$

$$V_P = \frac{R_1}{R_1 + R_2} V_o$$

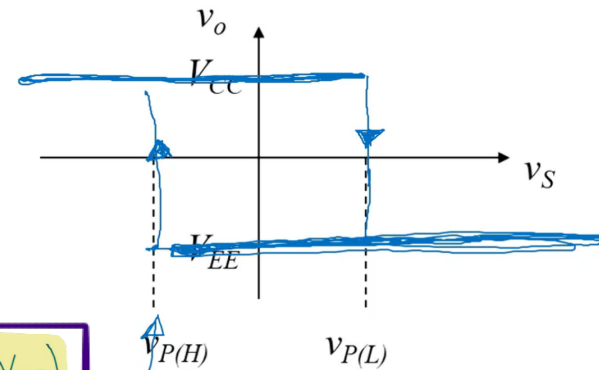
Output, $v_o = V_{CC} \quad (V_P > V_S)$
 $-V_{EE} \quad (V_P < V_S)$

$$V_S > \frac{R_1}{R_1 + R_2} V_{CC}$$

$$V_S > V_{P(H)}$$

$$V_{P(H)} = \frac{R_1}{R_1 + R_2} (-V_{EE})$$

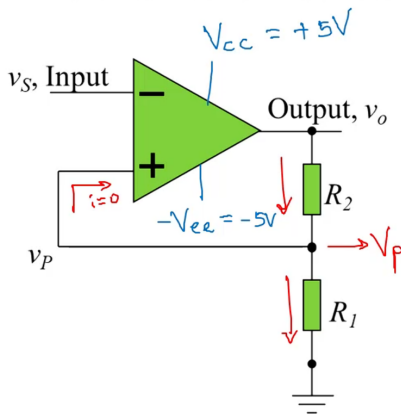
$$V_S < V_P$$



- This is an example of 'positive feedback'.
- Positive feedback leads to hysteresis.

Example 7.6

Calculate V_P when output hits rails. Assume $R_1 = R_2 = 1k\Omega$, and $V_{CC} = V_{EE} = 5V$



$$V_P = \frac{R_1}{R_1 + R_2} V_o$$

$$V_o = V_{CC} = +5V$$

$$V_{P(H)} = \frac{R_1}{R_1 + R_2} 5V = \frac{1k}{1k + 1k} 5V = 2.5V$$

Voltage needed to swing V_o from V_{CC} to V_{EE}

$$V_o = -V_{EE} = -5V$$

$$V_{P(L)} = \frac{1k}{1k + 1k} (-5) = -2.5V$$

Voltage needed to swing V_o from V_{EE} to V_{CC}

